

A MICRO-DOPPLER FREQUENCY ESTIMATION METHOD FOR MICRO-MOTION OBJECT USING TVAR MODEL

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Abstract

A micro-Doppler (mD) frequency estimation method for the precessional ballistic target is proposed in this paper. Based on the derived theoretical mD frequency model of the precessional ballistic target, we use the time-varying autoregressive (TVAR) method to estimate the frequency components contained in the returned signal and the results are rearranged to eliminate the association errors. Experimental results based on the electromagnetic computation data are presented, which demonstrates the algorithm performance under different SNR conditions and with the different target motion parameters.

Keywords: Micro-Doppler, Frequency estimation, Time-varying autoregressive (TVAR) model

1 Introduction

A typical micro-motion mode called precession will arise on the ballistic target (BT) during midcourse, which will induce additional frequency modulation on received radar signal, known as micro-Doppler (mD) effect^[1,2]. Study indicates that mD effect is highly related to the properties of target, which makes it a viable candidate for radar target classification and parameter estimation. When the target is precessing, the mD frequency is decided by its size, precession parameters and the relative position between radar and the target^[3]. Since the target is always composed of multiple scattering centers, the returned signal contains multiple time-varying frequency components and the mD frequency estimation of different scattering centers is vital for further work.

The returned radar signal from a target with micro-motion is nonstationary, and the time-frequency analysis methods for nonstationary signal can be categorized into nonparametric and parametric ones. The nonparametric approaches include short-time Fourier transform (STFT), Wigner-Ville Distribution (WVD), etc, and Viterbi algorithm^[4] or Markov model can be used to track the time-varying frequency components after acquiring the time-frequency distribution of the signal. Since there are drawbacks on the nonparametric time-frequency analysis, for example, the window effects and the low time-frequency resolution in STFT and the cross-term interference in WD, the performances of time-frequency distribution based estimation methods are inherently limited.

Parametric approaches are usually based on time-varying linear predictive models, these approaches describe the signal

using the time-varying model such as the autoregressive model (AR) or the autoregressive moving-average model (ARMA), which can achieve an improved accuracy of parameters estimation in comparison to other estimation methods^[5]. And when used as frequency estimator, these models show their advantages in describing nonstationary process especially when the data is short and the instantaneous frequency (IF) variation can be rapid or non-linear^[6].

In our work, a novel mD frequency estimation method for ballistic target using TVAR model is proposed. The remainder of this paper is organized as follows: In Section 2, expressions of the mD frequencies of different scattering centers are derived. The TVAR model is applied to frequency estimation in Section 3. The estimated results are rearranged in Section 4. More experimental results based on electromagnetic computation data are reported in Section 5. Section 6 concludes the paper by presenting some concluding remarks.

2 Micro-motion analysis of BT

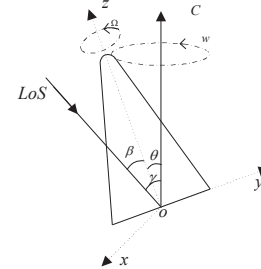


Figure 1 Model of Precession

As shown in Figure 1, $o-xyz$ is the target local coordinate system, and $\bar{z}$ is the symmetry axis of target, o is the center of bottom. Precession is defined as a target spinning around its axis of symmetry $\bar{o}z$ while rotating around the precession axis $\bar{o}C$ at the same time. The angle θ between $\bar{o}z$ and $\bar{o}C$ is called precession angle, the included angle of radar line of sight (LoS) and the precession axis $\bar{o}C$ is γ . When the target is precessing, the variation of included angle of LoS with the symmetry axis β can be expressed by Equation 1^[7], where w stands for the angular velocity of rotating, and φ_0 is the initial value of β when $t = 0$.

$$\cos \beta(t) = \cos \gamma \cos \theta + \sin \gamma \sin \theta \cos(wt + \varphi_0) \quad (1)$$

According to [8], when irradiated by radar, there are mainly three scattering centers on the surface of target, which are the nose of target P1, the bottom points P2 and P3. P2 and P3 are

the intersections of plane fixed by LoS and the symmetry axis with the bottom edge of the cone.

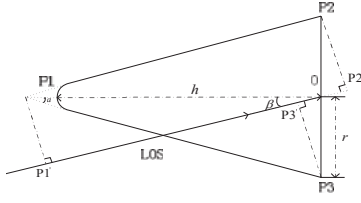


Figure 2 View of Scattering Centers

Assuming the distance from the origin of local coordinate system to radar is R_0 and translation has been fully compensated, then by projecting the scattering centers to the direction of LoS, we can get the distance variations of scattering centers to radar caused by precession. Let $P1'$, $P2'$ and $P3'$ denote the projection of $P1$, $P2$ and $P3$, and the corresponding distance $r_1(t)$, $r_2(t)$ and $r_3(t)$ can be expressed as follows, where h and r are labelled in Figure 2.

$$\begin{aligned} r_1(t) &= R_0 - h \cos \beta(t) \\ r_2(t) &= R_0 - r \sin \beta(t) \\ r_3(t) &= R_0 + r \sin \beta(t) \end{aligned} \quad (2)$$

The mD frequencies of the three scattering centers can be acquired by substituting the Equation (2) and Equation (1) to $f(t) = -\frac{2}{\lambda} \frac{dr(t)}{dt}$, and the frequency variations are:

$$f_1(t) = -\frac{2}{\lambda} \omega h \sin \theta \sin \gamma \sin(\omega t + \varphi_0) \quad (3)$$

$$f_2(t) = -\frac{2}{\lambda} r (\cos \theta \cos \gamma + \sin \theta \sin \gamma \cos(\omega t + \varphi_0)) \times \frac{\omega \sin(\omega t + \varphi_0) \sin \gamma \sin \theta}{\sqrt{1 - (\cos \gamma \cos \theta + \sin \gamma \sin \theta \cos(\omega t + \varphi_0))^2}} \quad (4)$$

$$f_3(t) = -\frac{2}{\lambda} r (\cos \theta \cos \gamma + \sin \theta \sin \gamma \cos(\omega t + \varphi_0)) \times \frac{\omega \sin(\omega t + \varphi_0) \sin \gamma \sin \theta}{\sqrt{1 - (\cos \gamma \cos \theta + \sin \gamma \sin \theta \cos(\omega t + \varphi_0))^2}} \quad (5)$$

When β varies from 0 to π , some scattering centers may not be seen on the radar sight, this phenomenon is called occlusion effect. The occlusion of scattering centers are shown in Table I, where a stands for the half-cone angle.

	$0 < \beta < a$	$\theta < \beta < \pi/2$	$\pi/2 < \beta < \pi - a$	$\pi - a < \beta < \pi$
P1	N	N	N	Y
P2	N	Y	N	N
P3	N	N	N	N

Table I Detail of Occlusion Effect N: Not Occluded. Y: Occluded.

We will use the electromagnetic computation data to verify the above analysis. Assume the height and radius of the target are 0.96m and 0.25m respectively, the carrier frequency of radar is 10.2GHz, pulse repetition frequency is 500Hz. Target's precession frequency $f_c = 2$ Hz while the precession angle $\theta = 10^\circ$. The occluded angles γ is set to 40° , suppose the initial value φ_0 is zero. Figure 4(a) and 4(b)

show the time-frequency distribution of the signal returned from the target and the theoretical mD frequency variations.

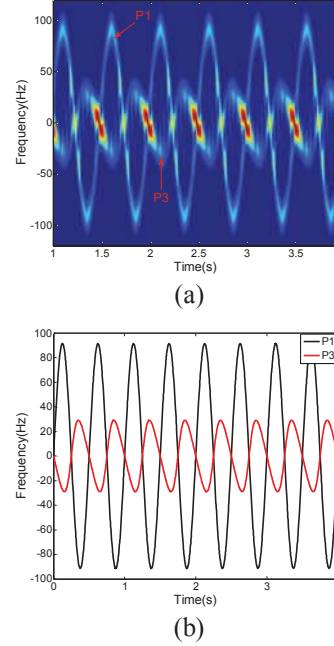


Figure 3 (a) Time-frequency distribution of returned signal. (b) Theoretical mD frequency variations

From the Figure 3 we can see that scattering center $P2$ is occluded by the target itself, and the electromagnetic computation result shows the same mD frequency variations as the theoretical analysis, all of these have proved the correctness of the above analysis.

3 Instantaneous frequency estimation using TVAR model

3.1 TVAR Model

A nonstationary process can be represented by Equation (6), which is referred to as TVAR model.

$$x(n) = -\sum_{k=1}^p a_k(n)x(n-k) + w(n) \quad (6)$$

where p is the model order, $w(n)$ is a zero-mean white Gaussian process with variance σ_w^2 , and $a_k(n)$ are TVAR coefficients which can be expressed by a summation of weighted time function:

$$a_k(n) = \sum_{m=0}^q a_{k,m} f_m(n) \quad (7)$$

$f_m(n)$ is called basis function, then Equation (6) can be expressed as:

$$x(n) = -\sum_{k=1}^p \sum_{m=0}^q a_{k,m} f_m(n)x(n-k) + w(n) \quad (8)$$

There are some available methods for the parameters estimation of TVAR model, such as Yule-Walker method, Burg method and modified approach. In this paper, the modified approach method is selected for its higher spectrum resolution

After acquiring the time-varying parameters, the instantaneous frequency (IF) of time-varying frequency

components at time n can be estimated by setting the denominator of Power Spectrum Intensity to zero and calculate the roots of the equation, since the IFs can be seen as the peak values of the signal's Power Spectrum Intensity $P(w, n)$.

$$P(w, n) = \frac{\sigma_w^2}{\left| 1 + \sum_{k=1}^p a_k(n) e^{-jw_k n} \right|^2} \quad (9)$$

Let $z = e^{-jw}$, rewrite the denominator as follow:

$$z^p + a_1(n)z^{p-1} + a_2(n)z^{p-2} + \dots + a_p(n) = 0 \quad (10)$$

Assume the roots are $z_k(n)$ $k=1,2,3,\dots,p$, and the IFs can be calculated by the following expression, where F_s is the sample rate.

$$f_k(n) = w_k(n) / 2\pi = \text{angle}(z_k(n)) * F_s / 2\pi \quad (11)$$

From (11) we can see that the denominator polynomial has p roots, therefore some roots may represent interference when the number of frequency components in the signal is less than p . In our work, we choose the roots that close to or on the unit circle of z -plane as the frequency components for the assumption $z = e^{-jw}$. Take a signal that contains two frequency components as example, when the model order $p = 5$, the roots of denominator polynomial are distributed as follow:

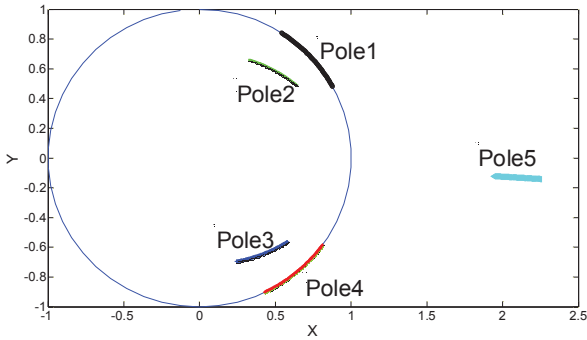


Figure 4 Distributions of poles on z -plane

From the Figure 4 we can see Pole 1 and 4 stand for frequency components in the signal while the remaining Poles represent interference or noise.

3.2 The application of TVAR model on electromagnetic computation data

The TVAR model will be tested on electromagnetic computation data for frequency estimation. In our problem, we choose the cosine basis function $f_m(n) = \cos(m\pi n/N)$, this is because the mD frequency variations are sinusoidal or quasi-sinusoidal. The same target and radar parameters are used here as those in Section II, and the model order p is set to 2 for noise is not considered here and only two scattering points are visible due to the occlusion effect, q is set to 50. The estimation and theoretical results are shown in Figure 5:

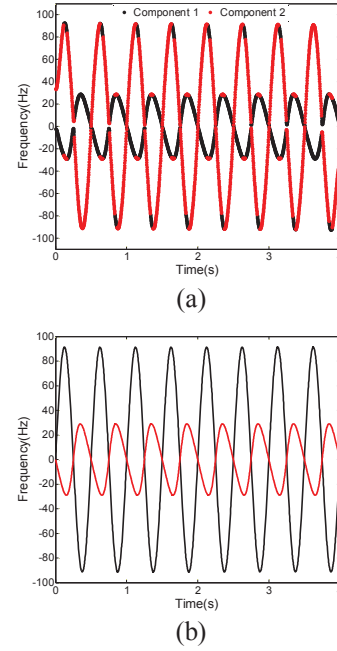


Figure 5 (a) Estimated results (b) Theoretical frequency variations

From Figure 5 (a) and (b), we can conclude that the estimated results match up with the theoretical results in dwell time, but there are multiple association errors on the results where two estimated values change their positions.

4 Estimated results rearrangement

Aiming at the problems arose in estimation, we will rearrange the estimated results in this section. Firstly we apply the Nearest Neighbour Algorithm (NNA) to the results, after that, the association errors occur only on the intersections of the two frequency components. The outputs of NNA are shown in Figure 6. From the theoretical analysis and experimental results we can see that the frequency variations near the intersections are approximately linear in a very short time, and the results will be rearranged by using this property.

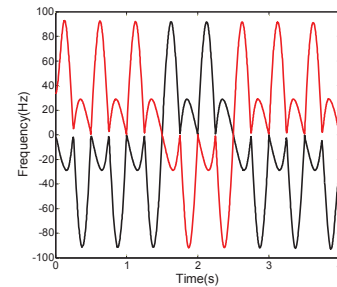


Figure 6 Outputs of NNA

Take the following intersectant area as example: assume the estimated results before the intersection are data1 and data2 while the estimated results after the intersection are data3 and data4, we can learn from the figure that after the linear fit, the slope of data1 is closer to the slope of data4 and that of data2 is closer to that of data3, which mean data1 and data4 belong to the same frequency component while data2 and data3

belong to another frequency component. Thus the association error in this area can be correct.

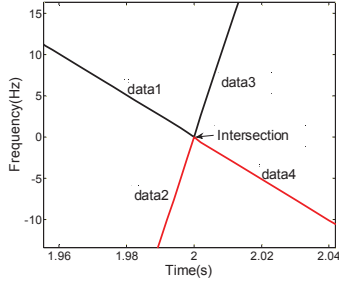


Figure 7 An intersectant area of the results
Based on the above analysis, the procedures of rearrangement are as follows:

1. Apply NNA to the original estimated results and find all intersections of frequency components, here we define the last moment of estimated results as the last intersection for convenience. Let N denotes the number of intersections, and $k = 1$.
2. Extract the data of the two components in a short time window τ before and after the k th intersection as data1 data2 data3 and data4 and calculate their slope after linear fit, rearrange the data according to the slope.
3. Apply the NNA again from the last moment of data3 and data4 to the $(k+1)$ th intersection.
4. $k = k + 1$, repeat procedure 2 and 3 until $k = N$, then all the frequency components will be rearranged.

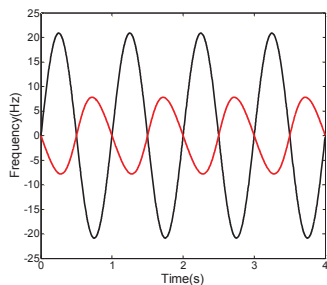
5 Experiments

In this section we will present more experimental results using the electromagnetic computation data to demonstrate the performance of the method proposed in this paper. The radar settings are the same as those in Section II. Two targets with different precession parameters are used here, their parameters are listed in Table II, here f_c stands for the precession frequency, θ is the precession angle, and γ is the occluded angle. The target 2 is the same target that used in section 3.2.

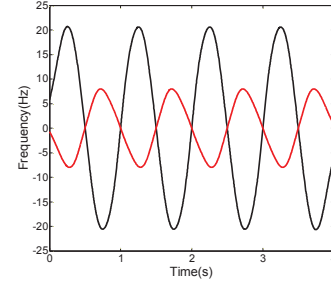
	f_c	θ	γ
Target 1	1Hz	5°	35°
Target 2	2Hz	10°	40°

Table II Motion Parameters

Figure 8 and 9 show the frequency estimated results after rearrangement and theoretical frequency variations of the returned signals, the signal-to-noise ratio (SNR) is set to 20dB.



(a) Theoretical mD frequency variations



(b) Estimated results: $p = 2$, $q = 30$

Figure 8 Theoretical and estimated frequency variations of target 1 (SNR=20dB)

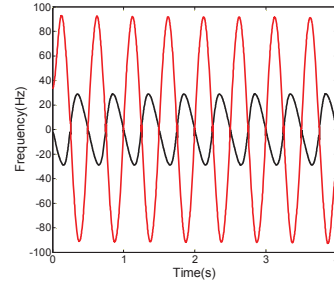
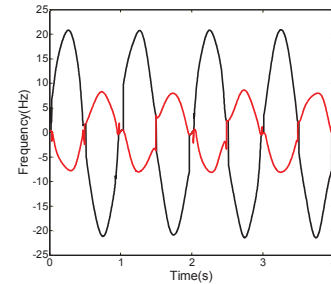


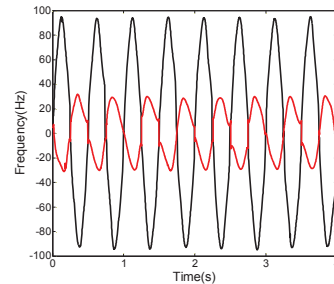
Figure 9 Estimated frequency variations of target 2 $p = 2$, $q = 50$ (SNR=20dB)

From Figure 8 and 9 we can learn that two frequency components have been correctly extracted from the signal, and the estimated results are consistent with the theoretical analysis, which have proved the effectiveness of the proposed method.

In practice, the robustness to noise is an important evaluation for the proposed algorithm, and the following two figures show the frequency estimated results when SNR is 5db:



(a) Target 1: $p = 7$, $q = 30$



(b) Target 2: $p = 6$, $q = 50$

Figure 10 Estimated results of two targets when SNR=5db
Comparison of Figure 10 and the theoretical analysis indicates that our estimated results are basically in

conformity with the theoretical analysis, which means the mD frequency still can be effectively estimated in the context of low SNR.

6 Conclusion

In this paper, we present a new micro-Doppler frequency estimation method for the precessional ballistic target. Firstly the micro-motion model is established and the theoretical expressions of the mD frequencies of different scattering centers are derived. Then the TVAR model is used to estimate the frequency components contained in the echo. In order to eliminate the association errors, the results are rearranged after the estimation. Experimental results based on the electromagnetic computation data demonstrate that the proposed method can estimated the micro-Doppler frequency of the precessional ballistic target effectively.

Acknowledgements

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